Detecting connected cars that send false messages

Cybersecurity for Connected Cars: Edge-Based Federated Intrusion Detection

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The problem

Connected cars broadcast their position several times a second over a direct radio link, and the cars around them brake and merge on it. Every message is signed, which proves which car sent it. The signature says nothing about whether the position is true, so a car with valid credentials can sign a false position and every cryptographic check still passes.

A receiver has one independent piece of evidence: how strongly the signal arrived, which depends on where the sender really is.

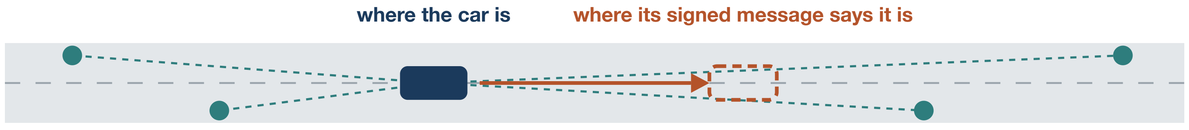


Figure 1. The attack. Receivers along the road (teal) measure how strongly each message arrives, which points back to the real car.

Our four aims

1. **One receiver.** Can a single receiver tell that a correctly signed position is false?
2. **Many receivers.** Can receivers that pool their measurements tell, and from what size of lie?
3. **A smart attacker.** Where would an informed attacker lie, and what stops it?
4. **Deployment.** Can roadside units learn the detector together without sharing raw data, and does it keep working when conditions change?

Table 1. What existing approaches check, and what they leave open.

Approach	Checks	Leaves open
Message signatures [5]	Which credential sent it	Whether the content is true
Plausibility checks [6]	Contradictions in one car's claims	A steady offset, which has none
Signal strength at one receiver [7]	Whether power fits the claimed distance	Fading, which looks the same
Cooperative checks [8]	One position, several receivers	The size of lie they can detect
Existing datasets [4]	Message content	Radio measurements
Federated detectors [9]	Training without sharing data	Published; we compare methods

What we built

We simulated a highway in the ns-3 network simulator with the 5G sidelink module, on which cars talk to each other directly [2, 3]. Honest cars carry realistic positioning error; attackers carry out ten kinds of misbehaviour, from false positions to flooding the channel. Every receiver records what each message claimed and what its radio measured. We found no public misbehaviour dataset over a 5G sidelink that records this.

6 km
of highway with realistic car following

7,916,708
labelled one-second windows [1]

10
kinds of attack, plus honest traffic

5
scenarios: traffic density, lie size, attacker timing, receiver placement

How often each attack is caught

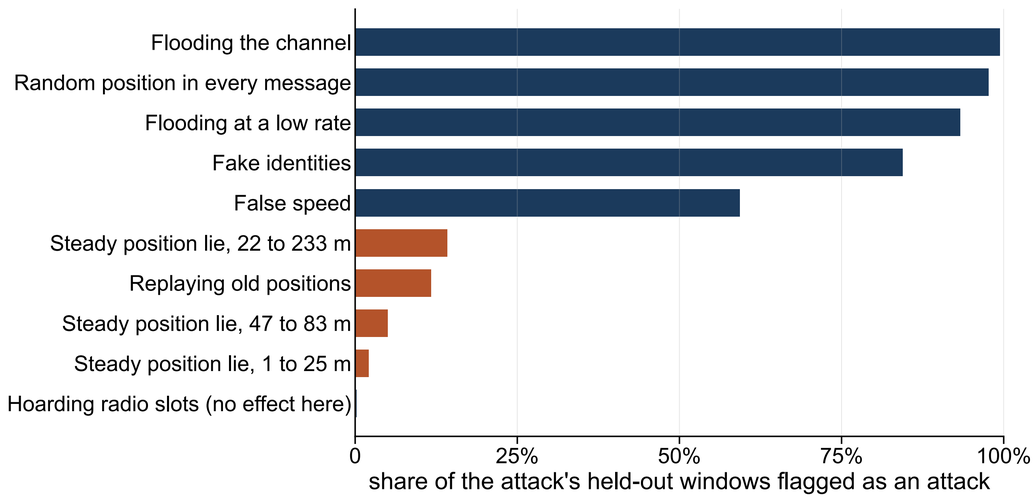


Figure 2. The baseline detector over all eleven classes scores 0.5319 on held-out cars (averaged detection score, 0 to 1). Flooding is caught almost always. Steady position lies and replays (rust) are mostly taken for honest traffic, which is the problem the rest of this work addresses.

One receiver against pooled receivers

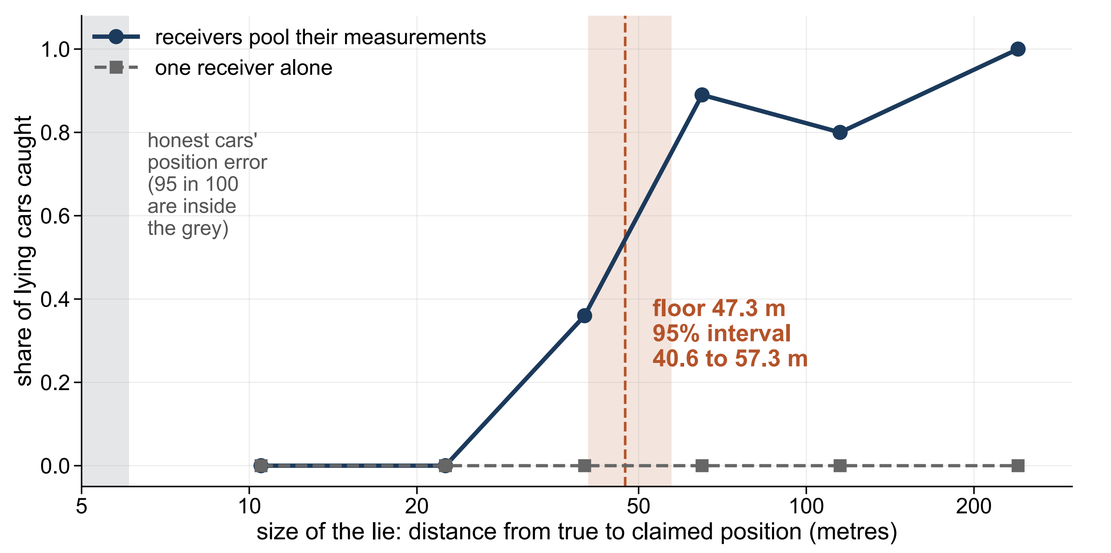


Figure 3. A single receiver catches no steady position lie at any size. Pooling received power across receivers catches half of all lies at **47.3 m** (95 percent interval 40.6 to 57.3 m), about two and a half times the 18.3 m error in locating a car from its signal. Pooling the receivers' verdicts instead catches none.

The weak direction and the map check



Figure 4. The receivers sit almost in a line along the road, so they measure position across it poorly. Their positions alone predict the attacker's best direction, 79.7° off the road, against 75 to 85° found by search. A lie in that direction leaves the road, where a map check rejects it.

Training across roadside units

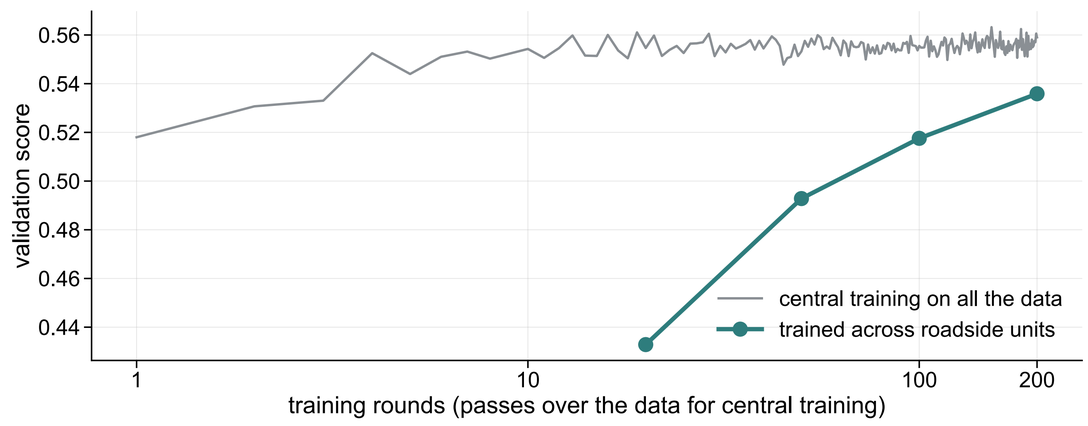


Figure 5. Trained across 96 roadside units without sharing raw data and scored on cars no unit had heard, the detector reaches 0.947 of central training on validation and 1.002 on test. Hiding each unit's update from the server costs no detection.

Conclusions against the aims

1. **One receiver** cannot detect a steady position lie at any size: content checks score 0.000 and the best of five learning methods 0.158 (Figure 3). The public VeReMi datasets show the same [4].
2. **Pooled receivers** detect it above 47.3 m, or 55.3 m when each roadside unit pools only its own area (Figure 3).
3. **The receivers' geometry** predicts where an informed attacker lies, and a map check covers that direction (Figure 4).
4. **Roadside units** can learn the detector together (Figure 5). After denser traffic or a weaker radio, refitting its input scaling on unlabelled local data recovers 77 and 87 percent of the loss; attackers lying in bursts cost only 0.0230 and refitting does not help.

Limitations and next steps

Every result comes from simulating one straight highway, and the radio model has not been checked against real measurements. The next step is to fit the same propagation model to real sidelink measurements, then test a junction and a curve, where the receivers no longer sit in a line.

Scan for the walkthrough and a live highway monitor, where every car is judged by a detector that never saw it. **cv2x-ids.vercel.app**

References

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